CSE 102 : Data Structure and Algorithms
End-Sem Lab Exam
Set 2 : MM 10

Instructions

- Students are supposed to attempt **only one of the following questions**
- Make sure that your code compiles. **Zero marks will be given if the code doesn't compile.**
- Name your cpp file as "**<roll No>.cpp**". Please submit only your .cpp files.
- Your final submission should be the fully executable code. You may/may not copy the sample code provided.
- We will be running all submissions through AI-sensitive Plag detection tools.

---

1. Cycle detection

For an undirected graph, the task is to check if there is a cycle in the given graph. You can use Vector from STL.

Input Format

- **Input n and m representing the number of nodes and the number of edges**
- **In the next m lines, take input for edges(space separated)**

5 5
1 0
0 2
2 1
0 3
3 4

Output

- **If the cycle exists, print True and all the connected nodes that make up the cycle. If multiple cycles exist, then print any one of them.**

True
0 1 2 0

- **If the cycle does not exist, print False.**

Example

-True
- 0 1 2 0

OR

2. Convert min heap to max heap

Given an array representation of min Heap, convert it to max Heap.
You can use vectors from STL.

Sample input

- **First line is the length of the array**
- **Second line contains the space separated elements of the array**

10
3 5 9 6 8 20 10 12 18 9

Sample output:

- **Output the final array after converting them into a max heap. Output n space separated integers.**

20 18 10 12 9 9 3 5 6 8

Constraints

- `1 <= n <= 300`
- All values are int only